Study notes: visualising molecules for photocatalyst DFT

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The main thing I learned: **a molecular picture is a measurement tool, not decoration.** Each kind of picture answers one question. A picture without a number next to it can mislead, because what you see depends on choices (viewing angle, isovalue, colour scale) that the molecule doesn't care about.

Level	Tool	Input	Question it answers
2D structure	RDKit	SMILES (graph only)	Is this the molecule and charge state I meant?
3D geometry	py3Dmol / RDKit	xyz coordinates	Is the shape sensible? Planar, twisted, crowded?
Before vs after	overlay	two xyz files	How far did DFT move the force-field structure?
Orbitals	PySCF → cube → viewer	MO coefficients	Where does the electron leave from and go to?

1 2D depictions: checking the input

A 2D drawing is computed from the bond graph alone, with no coordinates. It is the cheapest check in the whole workflow, and it catches the most expensive mistakes: a missing bromine, a neutral molecule that should be an ion, the wrong tautomer. These cost hours of DFT if they reach the next step.

Two RDKit tricks that turned out to be useful

- **SMARTS highlighting.** A SMARTS pattern is a search query for a substructure. Highlighting the xanthene core shows at once that eosin Y, fluorescein, rose bengal and rhodamine B share the same chromophore and differ only in what is attached to it. Writing the bonds as `~` ("any bond") makes the query match whatever bond orders the SMILES happens to use.
- **Atom indices.** When an error message or a script says "atom 12", draw the indices. The xyz files keep the same numbering, so this is also how I pick four atoms for a dihedral angle.

Takeaway

Look at the 2D picture of every molecule before any calculation. For Overleaf, export SVG (`fmt="svg"`) rather than PNG, so it stays sharp at any size.

2 3D geometry: is the shape sensible?

Coordinates come from RDKit: ETKDGv3 distance-geometry embedding followed by an MMFF94s force-field minimisation. The final version embeds 8 conformers per molecule (30 when there are

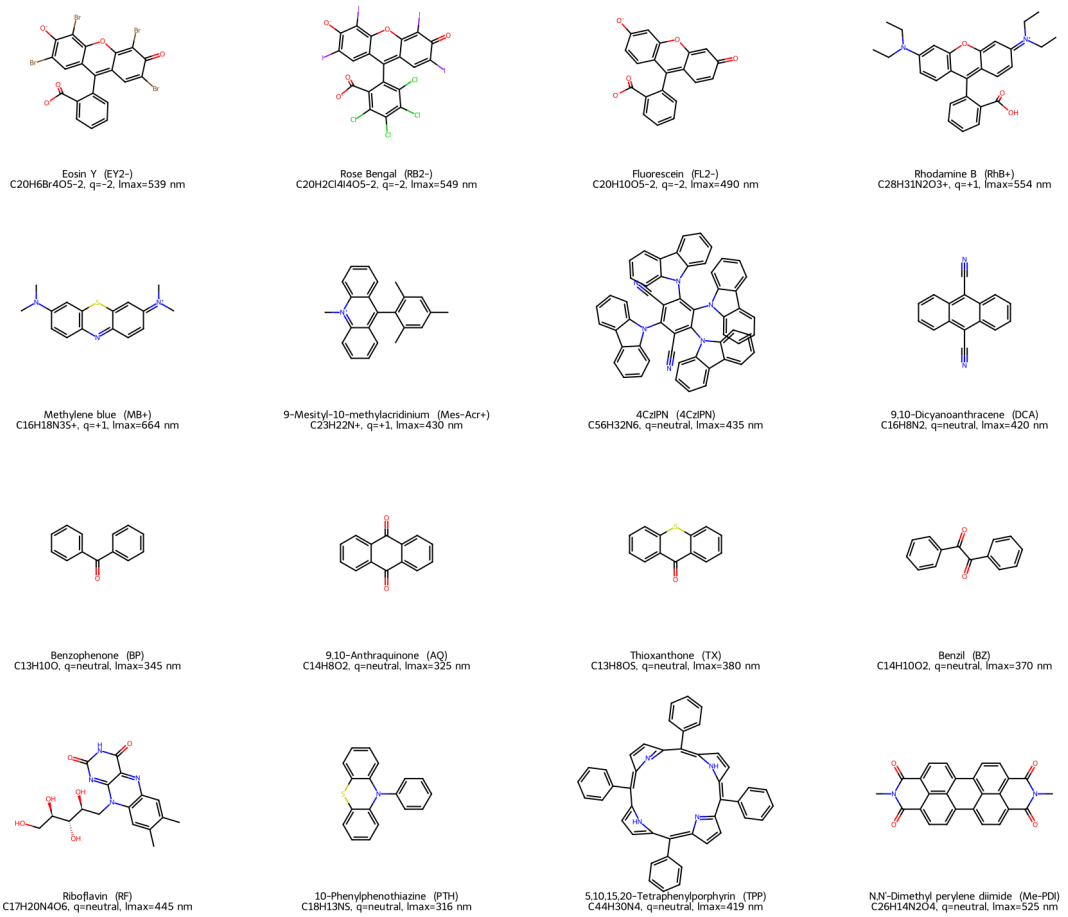


Figure 1: The 16 organic photocatalysts as encoded. The first thing to look at is the charges: EY^{2-}, RB^{2-} and FL^{2-} are dianions, and MB^{+}, RhB^{+} and Mes-Acr^{+} are cations. Those are the forms that absorb light under working conditions.

more than three rotatable bonds) and keeps the lowest-energy one. A single random embedding is fine for rigid dyes, but it can land in a poor local minimum for floppy ones.

Display styles each show something different

- **Ball-and-stick**: connectivity and angles. The default for checking a geometry.
- **Stick / line**: easier to read for large molecules such as TPP or 4CzIPN.
- **Space-filling**: atoms at van der Waals radii. Shows true size and crowding, and whether a molecule could lie flat on a surface or stack.

What the 3D pictures taught me about these catalysts Every one of these features is about *twisting*, and twisting controls conjugation. Two π systems held perpendicular can't share electrons, so they behave as separate units.

- **Eosin Y**: the benzoate ring is roughly perpendicular to the xanthene. I measured a ring–ring dihedral of 98.6° in my MMFF geometry. That's why the benzoate barely affects the colour: it is electronically cut off from the chromophore.
- **TPP**: the four *meso*-phenyls are twisted out of the porphyrin plane, so the absorbing π system is essentially the porphyrin macrocycle alone.

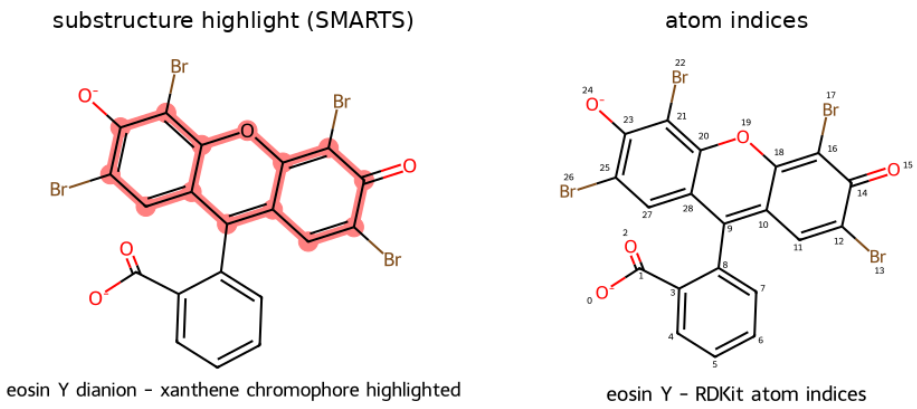


Figure 2: Eosin Y^{2-}: xanthene chromophore highlighted by SMARTS (left); RDKit atom indices (right).

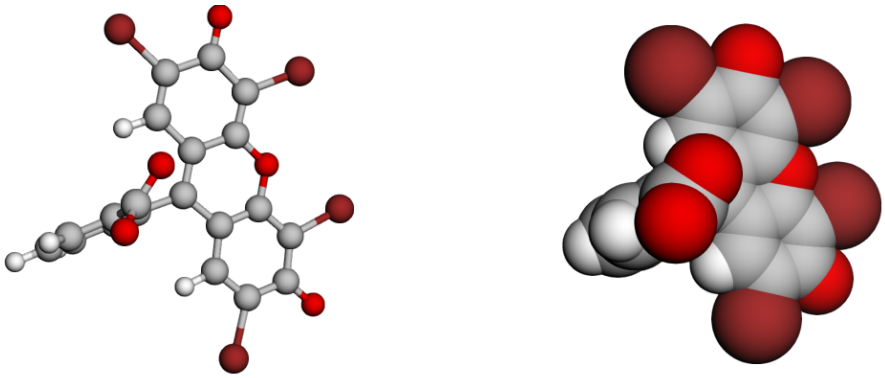


Figure 3: The same eosin Y^{2-} geometry in ball-and-stick (left) and space-filling (right). Ball-and-stick shows the benzoate ring seen edge-on. Space-filling shows how much of the molecule's volume the four bromines take up.

- **4CzIPN**: the four carbazoles are twisted out of the plane of the dicyanobenzene. That decoupling separates donor from acceptor, which is the design principle behind its thermally activated delayed fluorescence (TADF) and photoredox behaviour.
- **Mes-Acr$^+$**: the mesityl ring is held orthogonal to the acridinium by its two ortho-methyl groups. Same idea: a donor and an acceptor in one molecule but electronically apart, which is why its excited state is so strongly oxidising.
- **Riboflavin**: the flexible ribityl tail can take many shapes. One conformer is only a sample, which is a limitation to state.

Takeaway

Measure, don't eyeball. "Looks perpendicular" becomes a dihedral angle (`rdMolTransforms.GetDihedralDeg`). "Looks the same" becomes an RMSD. A number goes in the notes; an impression doesn't.

3 Overlays: how much did DFT change the structure?

Overlaying the force-field start with the DFT result is the honest way to decide whether the cheap geometry is good enough. Benzophenone was the instructive case:

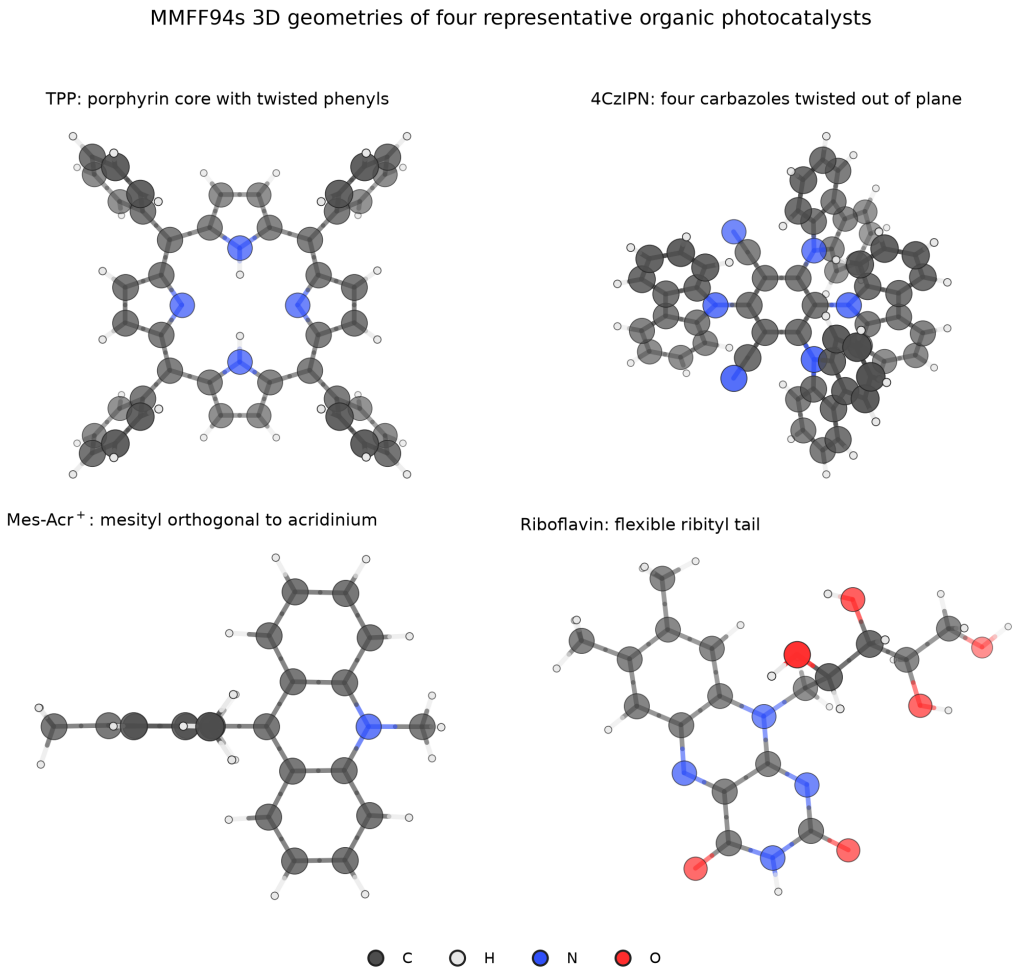


Figure 4: MMFF94s geometries of four photocatalysts, each showing one structural motif.

Molecule	$\mathrm{RMSD}_{\mathrm{heavy}}$ (Å)	Cycles	ΔE (kcal/mol)
anthraquinone (rigid)	0.019	15	−1.42
thioxanthone (rigid)	0.030	10	−1.88
benzophenone (2 rotatable rings)	0.096	11	−1.27

For benzophenone, DFT unwound the phenyl "propeller" from 35° to 29° and lengthened the C–$\mathrm{C}_{\mathrm{ipso}}$ bond from 1.485 to 1.501 Å.

Takeaway

- The viewing angle can hide the difference. Always look along at least two perpendicular directions.
- MMFF is very good for rigid fused rings ($\leq$0.03 Å) but noticeably worse for torsions between rings. Errors concentrate exactly where conjugation is decided.
- So validating MMFF on three rigid molecules does *not* justify using it for the floppy or charged ones (rhodamine B, riboflavin, eosin Y^{2-}). Those still need a DFT optimisation, or the limitation stated.

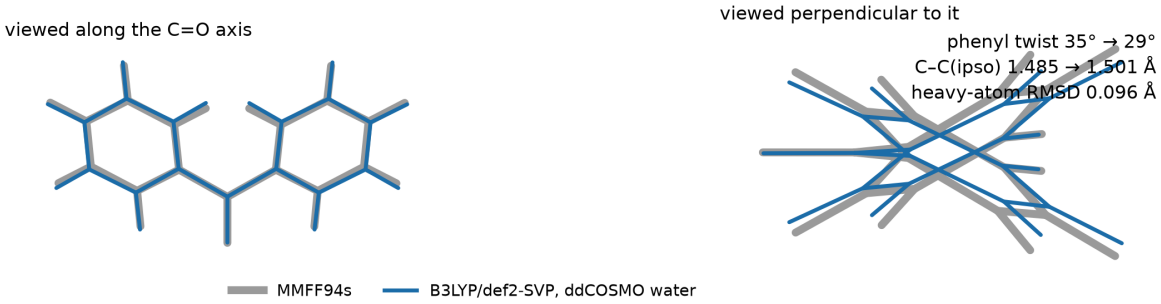


Figure 5: MMFF94s (grey) vs B3LYP/def2-SVP in ddCOSMO water (blue). The right-hand view makes the change in twist visible; the left-hand view hides it completely.

4 Orbitals: where are the electrons?

How the picture is made. PySCF evaluates an orbital on a 3D grid and writes a **Gaussian cube file**. That's a plain-text format that py3Dmol, VESTA, VMD, Avogadro and Jmol all read. An orbital has a sign, so it is drawn as two isosurfaces, +iso and −iso, in two colours.

- Where the colour flips there is a **node**: a surface where the orbital is zero.
- The colours are arbitrary. Swapping blue and red gives the same orbital, since ψ and $-\psi$ are physically identical.
- The **isovalue is a choice** (0.02–0.05 a.u. is conventional). Use the same value for every orbital you compare.

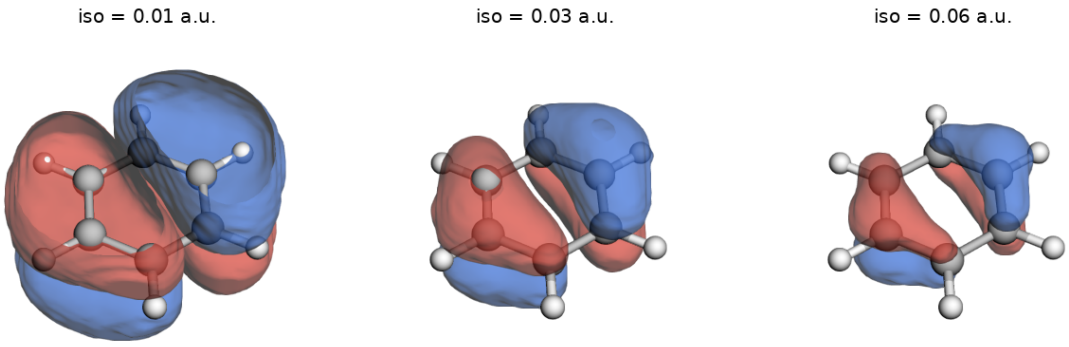


Figure 6: The same benzene HOMO at three isovalues. The lobes shrink and swell, but the *nodes stay put*. The nodal pattern is the physics; the lobe size is a display choice.

Benzene: the textbook check.

- HOMO and HOMO−1 have the same energy (−6.97 eV). They are a degenerate pair (e_{1g} in D_{6h}), each with one nodal plane through the ring. The orientation the program prints is arbitrary, since any mixture of the two is equally valid. **Never interpret one member of a degenerate pair on its own.**
- The LUMO has two nodal planes. More nodes means higher energy: the particle-on-a-ring picture, seen directly.

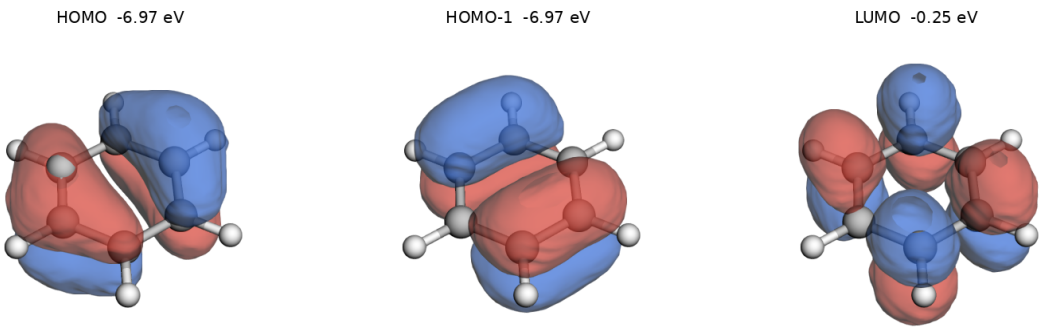


Figure 7: Benzene B3LYP/def2-SVP frontier orbitals, ±0.03 a.u.

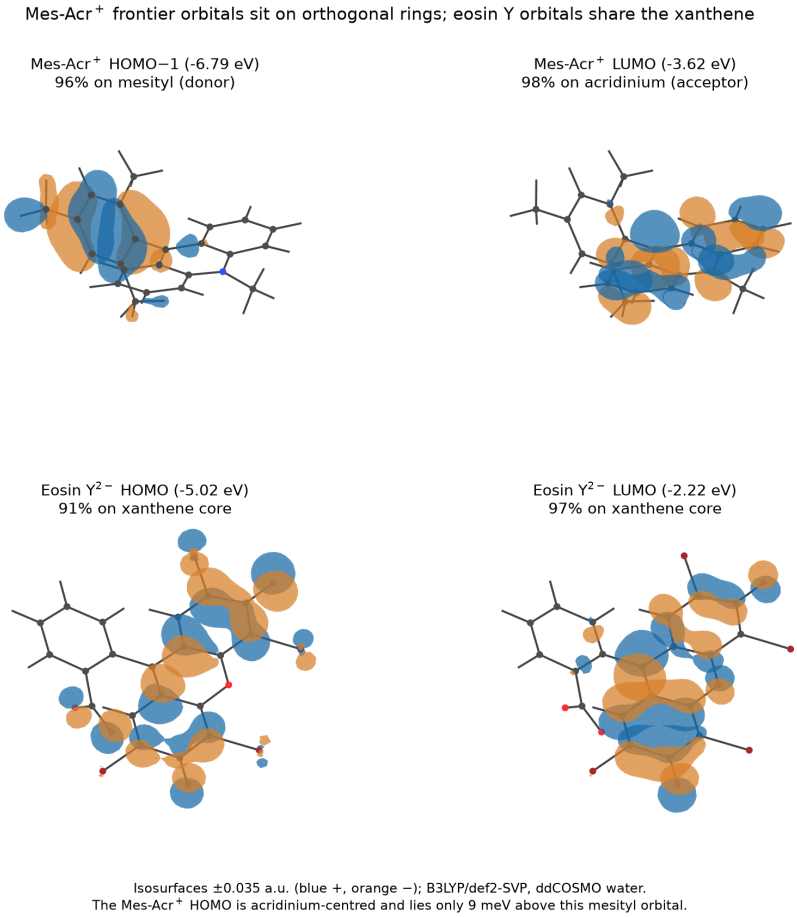


Figure 8: Mes-Acr$^+$ (top) and eosin Y^{2-} (bottom), B3LYP/def2-SVP in ddCOSMO water, ±0.035 a.u.

Photocatalysts: local vs charge-transfer excitation. Pictures alone are not enough here, so each orbital was also split into fragment percentages (Mulliken populations):

Molecule	Orbital	E (eV)	Where it lives
Eosin Y^{2-}	HOMO	−5.02	91% xanthene core
	LUMO	−2.22	97% xanthene core
Mes-Acr$^+$	HOMO−2	−6.80	>99% mesityl
	HOMO−1	−6.79	96% mesityl
	HOMO	−6.79	99% acridinium
	LUMO	−3.62	98% acridinium

- **Eosin Y**: HOMO and LUMO both sit on the xanthene, so the lowest excitation is a *local* $\pi \rightarrow \pi^*$ excitation.
- **Mes-Acr$^+$**: the LUMO is on the acridinium acceptor, and mesityl donor orbitals sit just below the HOMO. Exciting from mesityl to acridinium moves charge between two orthogonal rings: a *charge-transfer* state. This links directly back to the 3D picture in Section 2.
- The three top occupied orbitals of Mes-Acr$^+$ lie within 13 meV of each other, which is far below the accuracy of B3LYP. **Which one is "the HOMO" is not meaningful here**, so look at the set, not the label.

Takeaway

Pair every orbital picture with a number: the fragment percentage, the energy, and how close it is to its neighbours. Mulliken percentages depend on the basis set, so treat them as fragment-level indicators (91% vs 10%), not precise values. For charge-transfer molecules (Mes-Acr$^+$, 4CzIPN), B3LYP underestimates the excitation energy; use a range-separated functional (CAM-B3LYP, ωB97X-D) for TDDFT.

5 Energy-level diagrams: the picture that misleads most easily

Plotting HOMO/LUMO for all 16 molecules side by side is a good way to see trends. The Kohn–Sham gap tracks the lowest experimental absorption band well ($r = 0.90$, MAE 0.45 eV, $n = 16$). The biggest outlier, benzophenone (+1.37 eV), is explained by its weak first band being n$\rightarrow \pi^*$, not $\pi \rightarrow \pi^*$.

Pitfall

Drawing the H^+/H_2 and O_2/H_2O levels across Kohn–Sham orbital energies invites the reader to read off redox driving forces, and the numbers don't support that. As I found with benzene, $-\varepsilon_{\mathrm{HOMO}}$ underestimates the ionisation energy by about 2 eV for B3LYP, and the HOMO–LUMO gap is neither the fundamental nor the optical gap. A good correlation with absorption shows the *trend* is right, not that the absolute positions are. For redox potentials, compute energy differences between charge states in solvent (ΔSCF); for absorption, use TDDFT.

6 Practical notes on tools

- **RDKit**: identifiers, 2D drawing, conformers, xyz export, dihedrals and RMSD.
- **py3Dmol**: the quickest interactive viewer in Jupyter, and it writes standalone HTML I can send to my PI. It has no headless PNG export, so static figures need a workaround. I used a headless browser (Playwright + Chromium); the Claude Science run used marching cubes in matplotlib. matplotlib only approximates which surface is in front, so treat those as schematic.
- **VESTA / VMD**: what to use for figures with proper lighting and depth. VESTA also handles crystal structures, which I'll need in Week 2.
- **Cube files** are 3–5 MB each. Keep them out of git (`*.cube` in `.gitignore`) and regenerate them, which takes seconds once the SCF is done.

How I did it

- 2D: `viz.draw_2d(smiles, highlight_smarts=..., atom_indices=True)`
- 3D: `viz.view_xyz(xyz, "ballstick" | "sphere" | "stick")`
- Overlay: `viz.overlay_xyz(xyz_mmff, xyz_dft)`
- Orbital: `cubegen.orbital(mol, "x.cube", mf.mo_coeff[:, i])`, then `viz.view_orbital("x.cube", iso=0.03)`
- Everything is in `molecules/visualize.ipynb` (or the self-contained `visualize_molecules.ipynb`).

7 Checklist I'll reuse

1. 2D picture of every input: right atoms, right charge, right protonation state?
2. 3D picture: rings flat where they should be, twisted where sterics demand it? Measure the key dihedrals.
3. Overlay MMFF vs DFT from two viewing directions; record heavy-atom RMSD.
4. Orbitals: same isovalue for everything; check for degenerate or near-degenerate partners before naming "the HOMO".
5. Put a number next to every picture (energy, % localisation, dihedral, RMSD).
6. Never read redox potentials or absorption energies off Kohn–Sham orbital energies.
7. Label every figure with its level of theory (functional/basis/solvent). These runs used B3LYP/def2-SVP in ddCOSMO water, with *no* D3 dispersion.